

Workshop 06: Actuator Example



Overview

- **Actuator Example**
 - This workshop will discuss how to use Maxwell in order to develop a complete analysis of a simple actuator in 2D Cylindrical symmetry
 - The following tasks will be performed:
 - Create the actuator model
 - Perform a Magnetostatic Analysis
 - Perform a Parametric Magnetostatic Analysis
 - Perform a Transient analysis
 - Analyze results

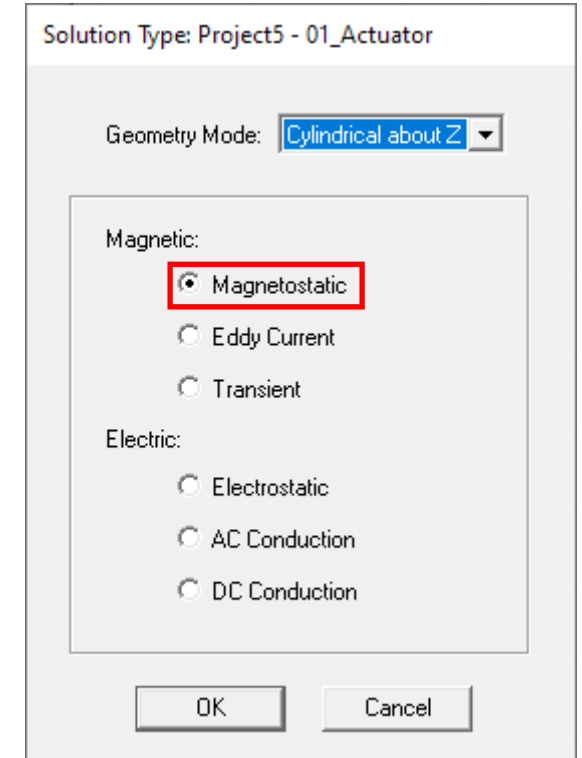
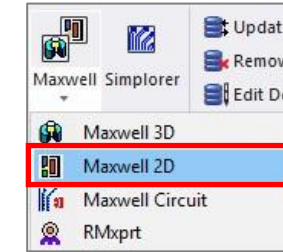
Model Setup

- Insert Design

- Select the menu item *Project* → *Insert Maxwell 2D Design*, or click on the icon in drop down list Maxwell on panel Desktop
- Name the Design “01_Actuator”

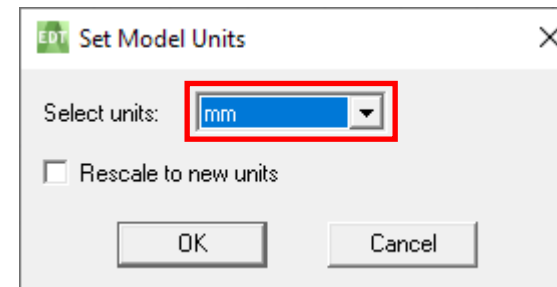
- Set Solution Type

- Select the menu item *Maxwell 2D* → *Solution Type*
- Geometry Mode: *Cylindrical about Z*
- Choose *Magnetic* → *Magnetostatic*
- Click the OK button



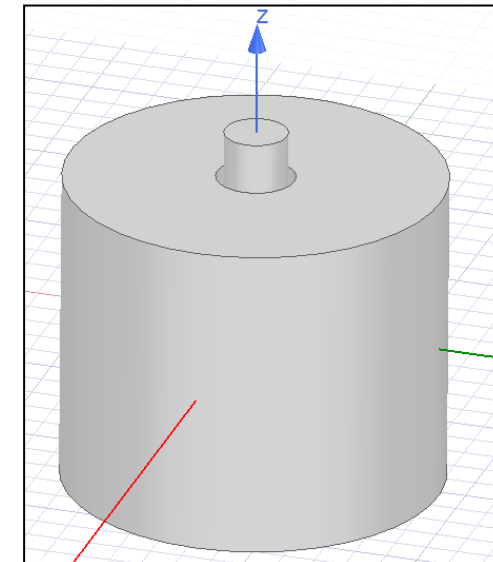
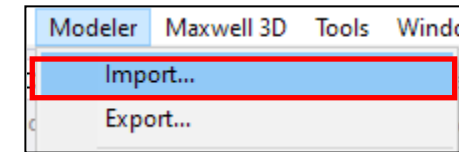
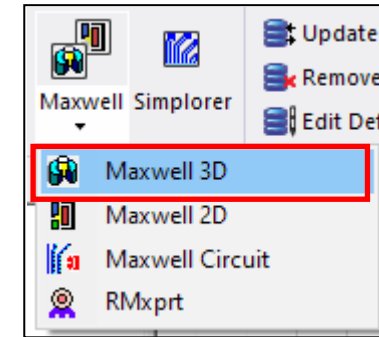
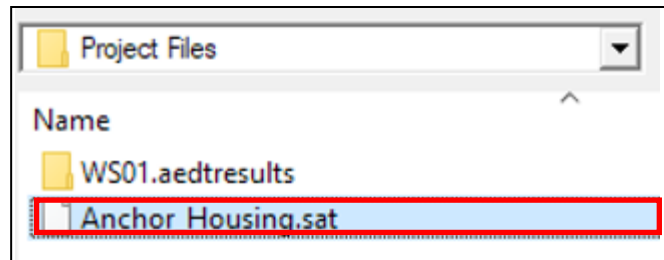
- Set Model Units

- Select the menu item *Modeler* → *Units*
- In Set Modeler Units window,
 - Select units: *mm* (millimeters)
 - Press the OK button



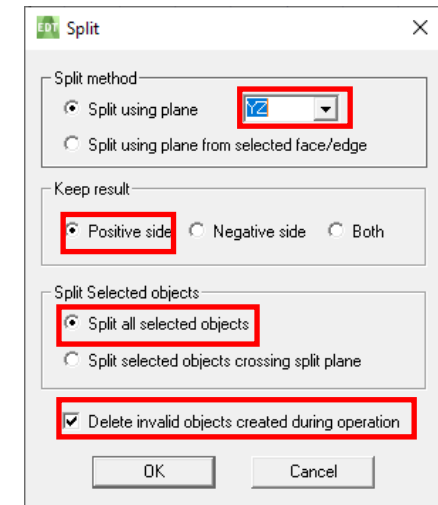
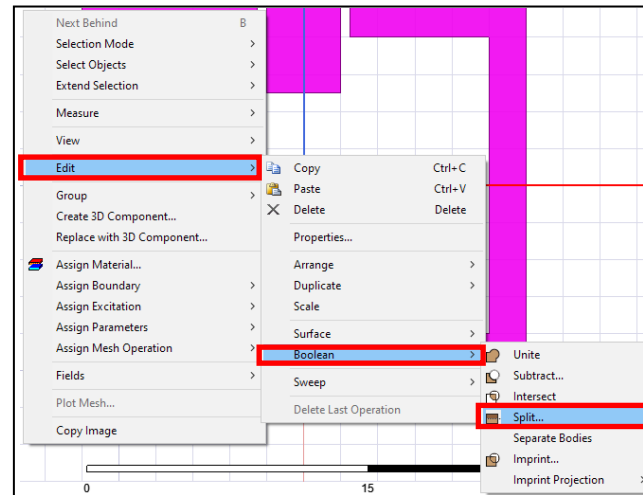
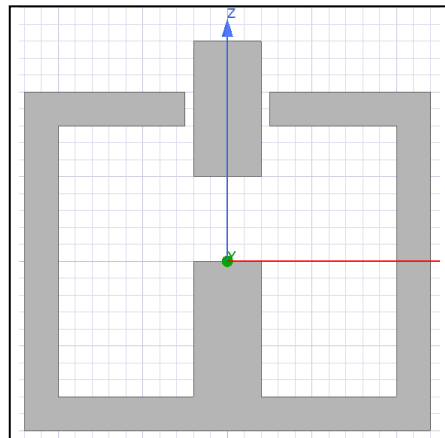
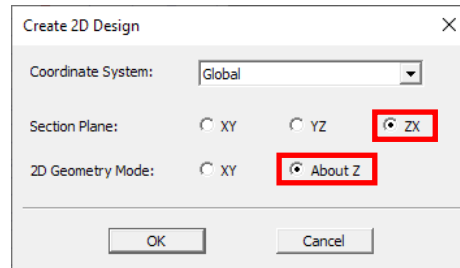
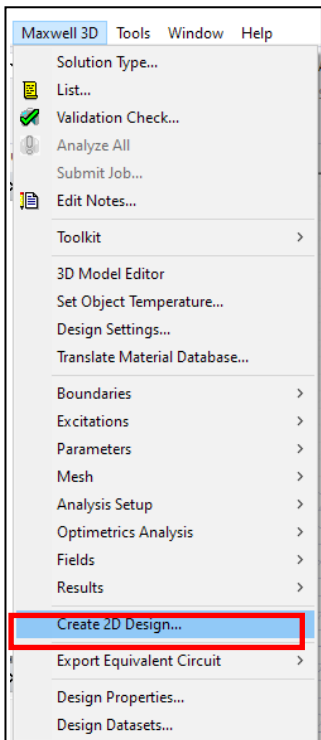
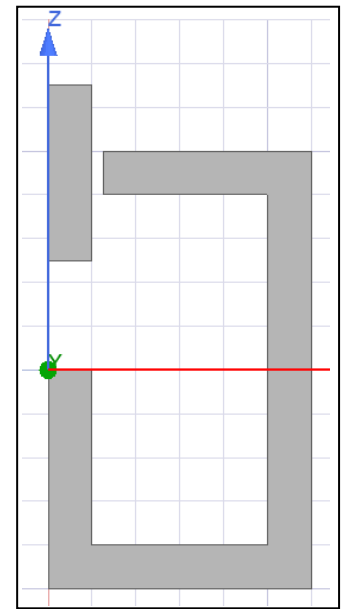
Create Model

- Create the Housing and Anchor
 - There are two ways of creating geometries in Maxwell
 - Importing .sat files or other CAD files to Maxwell
 - Drawing geometries from scratch
 - Creating Housing and Anchor by importing .sat file
 - Select the menu item **Project** → **Insert Maxwell 3D Design**, or click on the icon in drop down list Maxwell on panel Desktop
 - Select the menu item **Modeler** → **Import**
 - Choose **Anchor_Housing.sat** → open



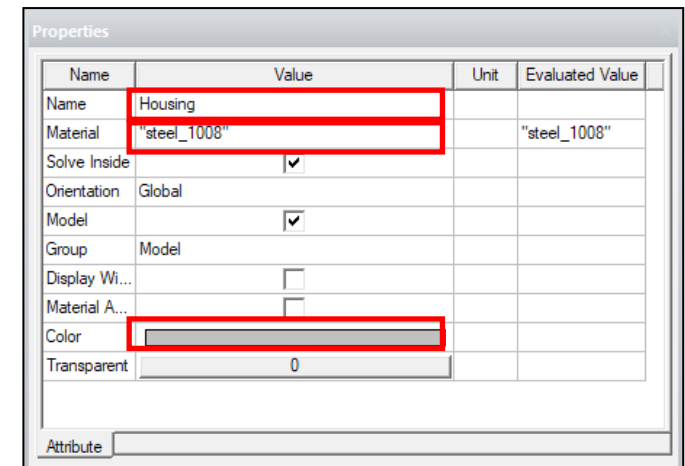
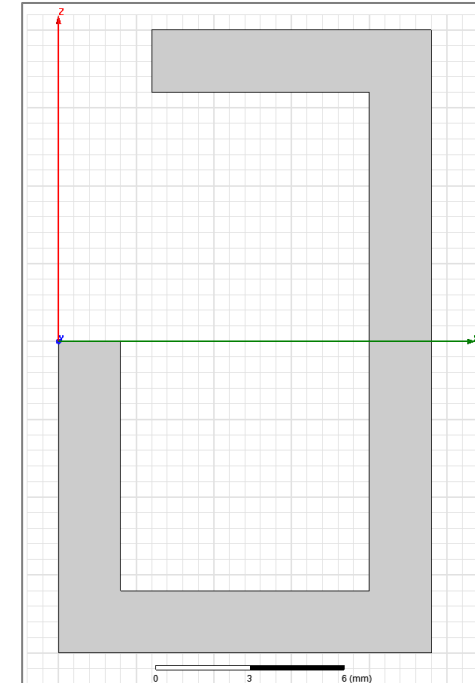
Create Model

- Create the Housing and Anchor
 - Select the menu item **Maxwell** → **Create 2D Designs**
 - Select **ZX** for “Section Plane”, and **About Z** for “2D Geometry Model”
 - Select both Housing and Anchor, **right click** → **Edit** → **Boolean** → **Split**
 - Select **YZ** for “Split using plane”, keep result for **Positive side**, choose **Split all selected objects**, and check **Delete invalid objects created during operation**



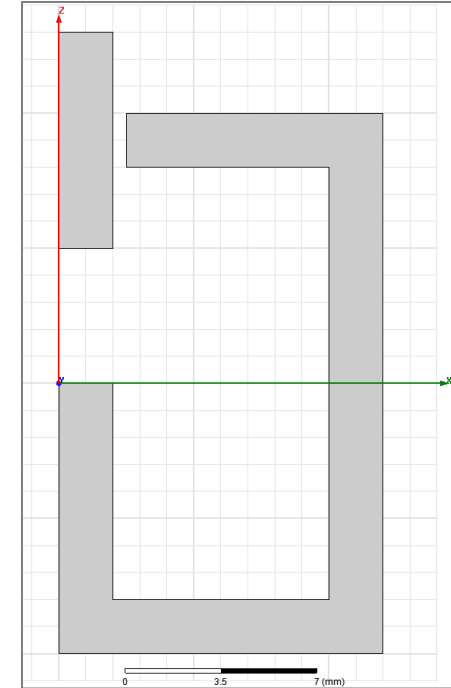
Create Model

- Create the Housing
 - Select the menu item **Draw** → **Line**
 - Using the coordinate entry fields, enter the following points
 - X: 0, Y: 0, Z: 0, *Press the **Enter** key*
 - X: 0, Y: 0, Z: -10, *Press the **Enter** key*
 - X: 12, Y: 0, Z: -10, *Press the **Enter** key*
 - X: 12, Y: 0, Z: 10, *Press the **Enter** key*
 - X: 2.5, Y: 0, Z: 10, *Press the **Enter** key*
 - X: 2.5, Y: 0, Z: 8, *Press the **Enter** key*
 - X: 10, Y: 0, Z: 8, *Press the **Enter** key*
 - X: 10, Y: 0, Z: -8, *Press the **Enter** key*
 - X: 2, Y: 0, Z: -8, *Press the **Enter** key*
 - X: 2, Y: 0, Z: 0, *Press the **Enter** key*
 - X: 0, Y: 0, Z: 0, *Press the **Enter** key twice*
 - Change the name of resulting sheet to **Housing** and color to **Gray** using the property window
 - Change the material of the sheet to **Steel_1008**



Create Model

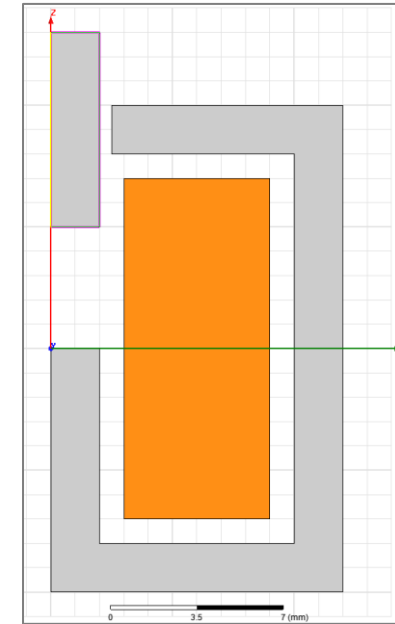
- Create the Anchor
 - Select the menu item **Draw** → **Rectangle**
 - Using the coordinate entry fields, enter the position of rectangle
X: 0, Y: 0, Z: 13, Press the **Enter** key
 - Using the coordinate entry fields, enter the opposite corner
dX: 2, dY: 0, dZ: -8, Press the **Enter** key
 - Change the name of resulting sheet to **Anchor** and color to **Gray** using the property window
 - Change the material of the sheet to **Steel_1008**



Properties			
Name	Value	Unit	Evaluated Value
Name	Anchor		
Material	"steel_1008"		"steel_1008"
Solve Inside	☒		
Orientation	Global		
Model	☒		
Group	Model		
Display Wi...	☐		
Material A...	☐		
Color			
Transparent	☐ 0		
Attribute			

Create Model

- Create the Coil
 - Select the menu item **Draw** → **Rectangle**
 - Using the coordinate entry fields, enter the position of rectangle
X: 3, Y: 0, Z: 7, Press the Enter key
 - Using the coordinate entry fields, enter the opposite corner
dX: 6, dY: 0, dZ: -14, Press the Enter key
 - Change the name of resulting sheet to **Coil** and color to **Orange** using the property window
 - Change the material of the sheet to **Copper**



Name	Value	Unit	Evaluated Value
Name	Coil		
Material	"copper"		"copper"
Solve Inside	☒		
Orientation	Global		
Model	☒		
Group	Model		
Display Wi...	☐		
Material A...	☐		
Color			
Transparent	0		

Attribute

Define Region

- Create Simulation Region

- Select the menu item **Draw** → **Region** or click on the icon



- In Region window

- Pad all directions similarly: ☒ **Checked**
 - Padding Type: Percentage Offset
 - Value: **100**
 - Press **OK**

Note: the region extension in negative X direction is neglected due to RZ-symmetry about the Z-axis.

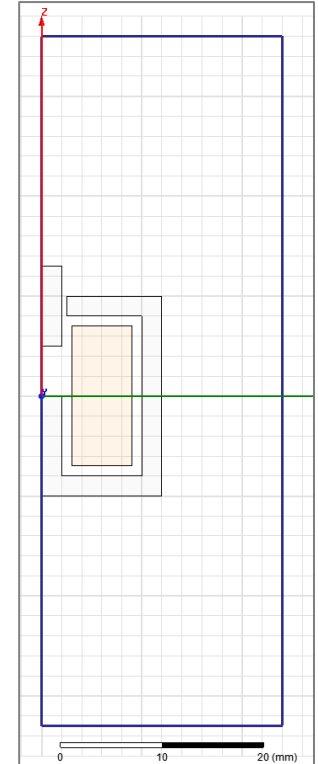
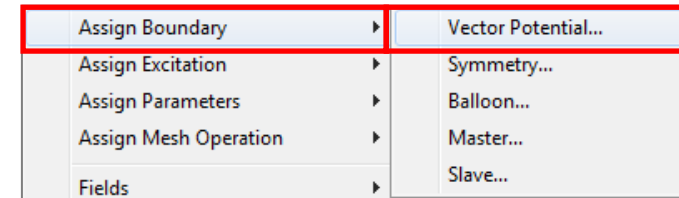
100 padding percentage is chosen to define the region size which should sufficiently capture the fringing flux, in certain cases, a larger sized region may be required.

Direction	Padding type	Value	Units
All	Percentage Offset	100	

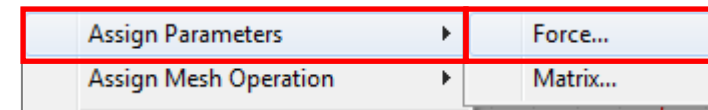
Assign Boundary Conditions and Force Parameter

- Assign Boundary Conditions to Region Edges
 - Select the menu item **Edit** → **Select Mode** → **Edges**
 - Select the three edges of the Region
 - Select the menu item **Maxwell 2D** → **Boundaries** → **Assign** → **Vector Potential**
 - In Vector Potential Boundary window, Press OK

Note: the choice of vector potential boundary condition confines the magnetic flux to the size of the region



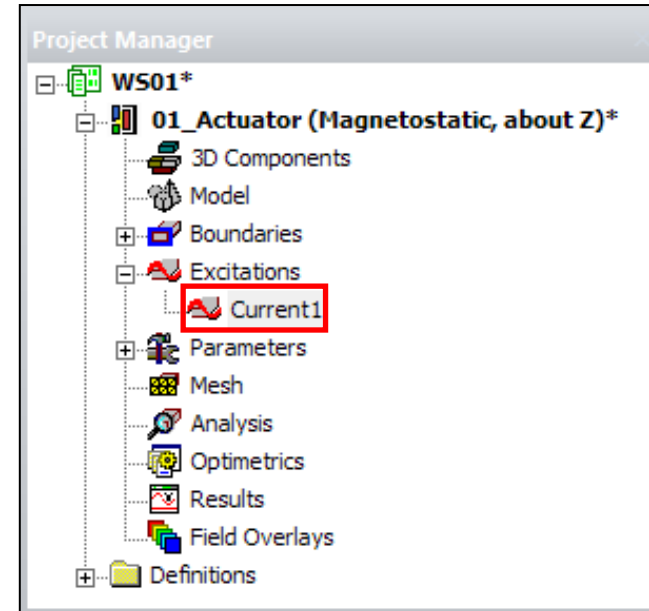
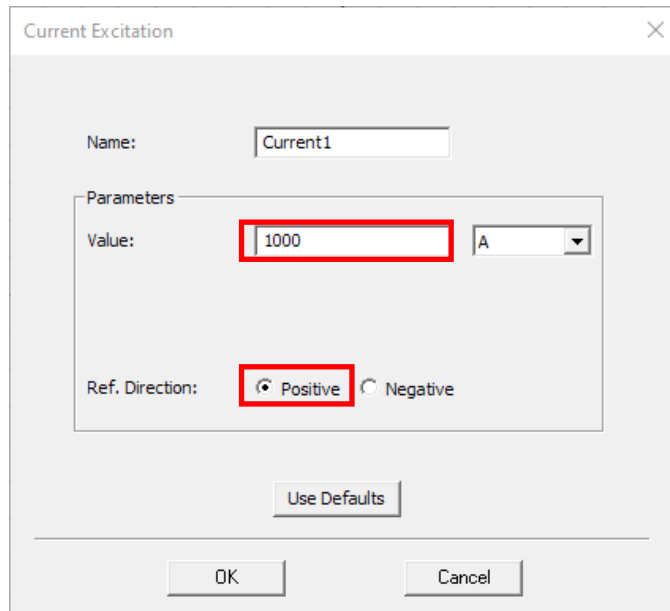
- Assign Force Calculation
 - Select the object **Anchor** from history tree
 - Select the menu item **Maxwell 2D** → **Parameters** → **Assign** → **Force**
 - In Force Setup window, Press OK



/ Assign Excitations

- Assign Excitations
 - Select the sheet **Coil** from history tree
 - Select the menu item **Maxwell 2D → Excitations → Assign → Current**
 - Name: **Current1**
 - Value: **1000**
 - Ref. Direction: **Positive**
 - In the Project Manager window under Excitations a new item called “Current1” is present

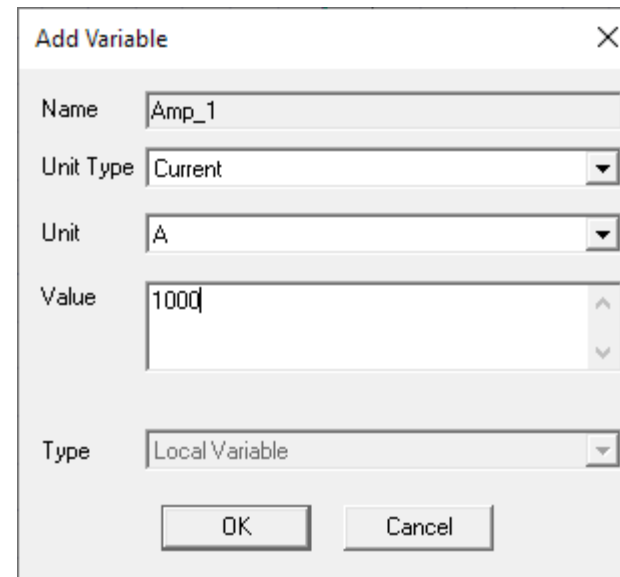
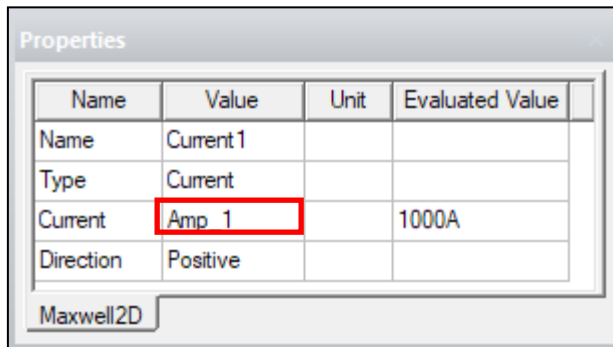
Note: A value of 1000 represents 1000 A-turns



/ Prepare the Parametric analysis

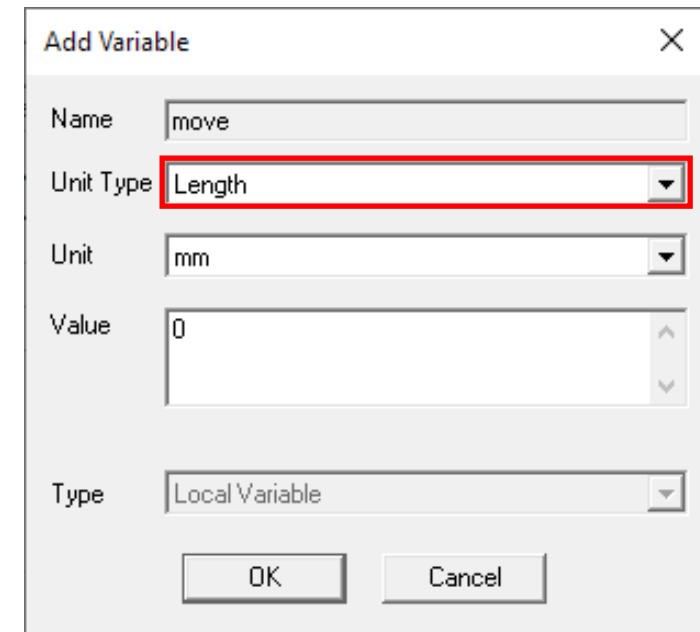
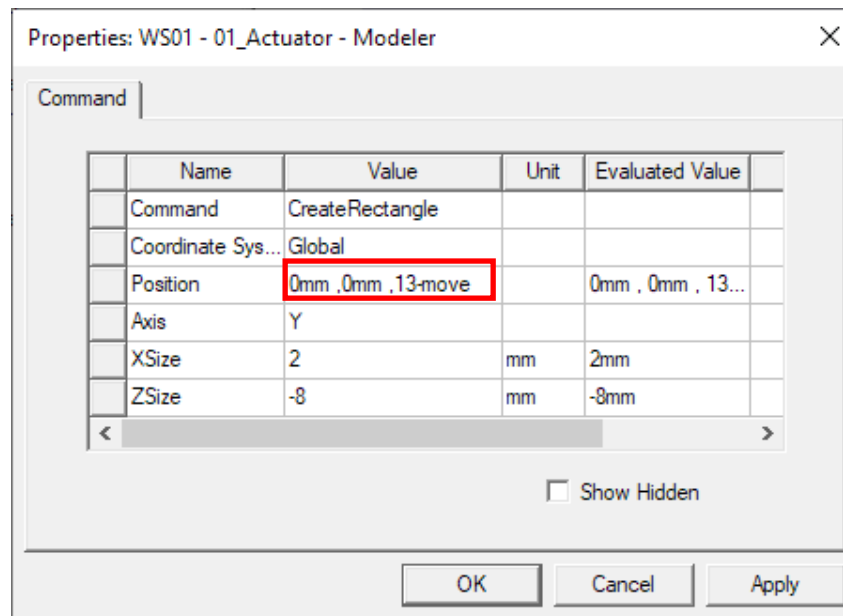
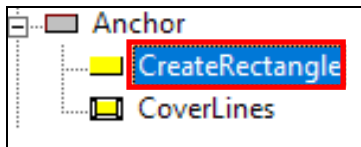
- **Modify Excitations**

- Select the just created Excitation **Current1**
- In the property window, change the current value from 1000 to **Amp_1**
- The **Add Variable window** pops-up with all the fields already fulfilled. Press Ok
- The new local variable **Amp_1** has been added to 01_Actuator Design



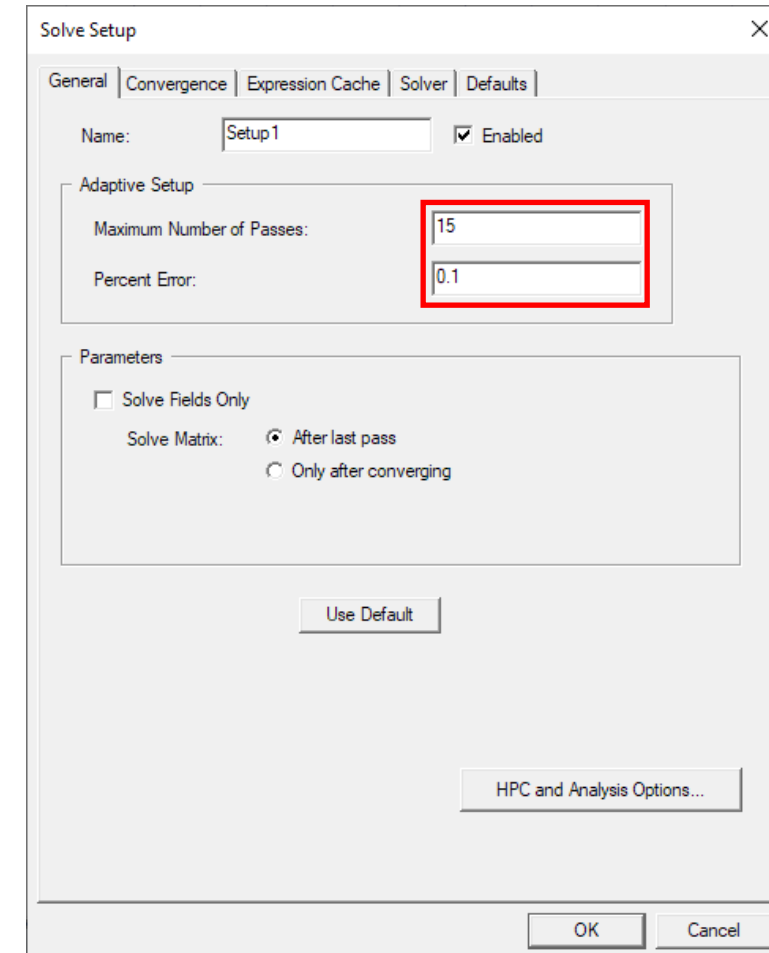
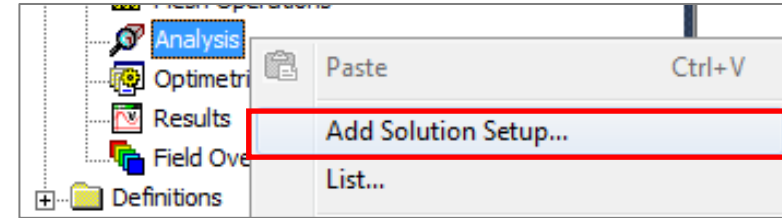
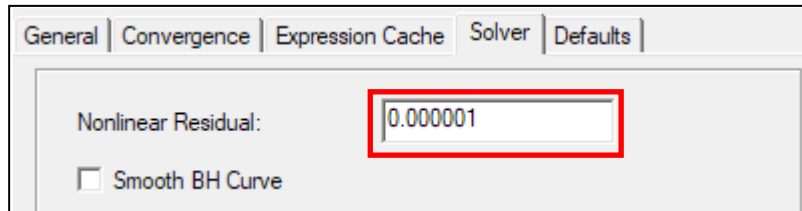
Prepare the Parametric analysis

- **Modify Anchor Position**
 - Select **CreateRectangle** from **Anchor** from the history tree
 - In the property window, change the Z-Position from 13 mm to **13mm - move**
 - The **Add Variable window** pops-up.
 - Unit Type: Length
 - Press OK
 - The new local variable **move** has been added to 01_Actuator Design



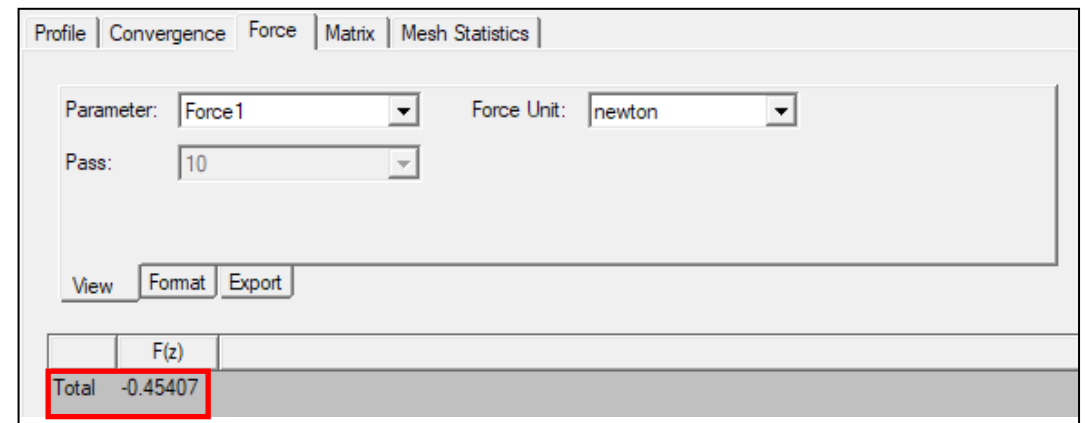
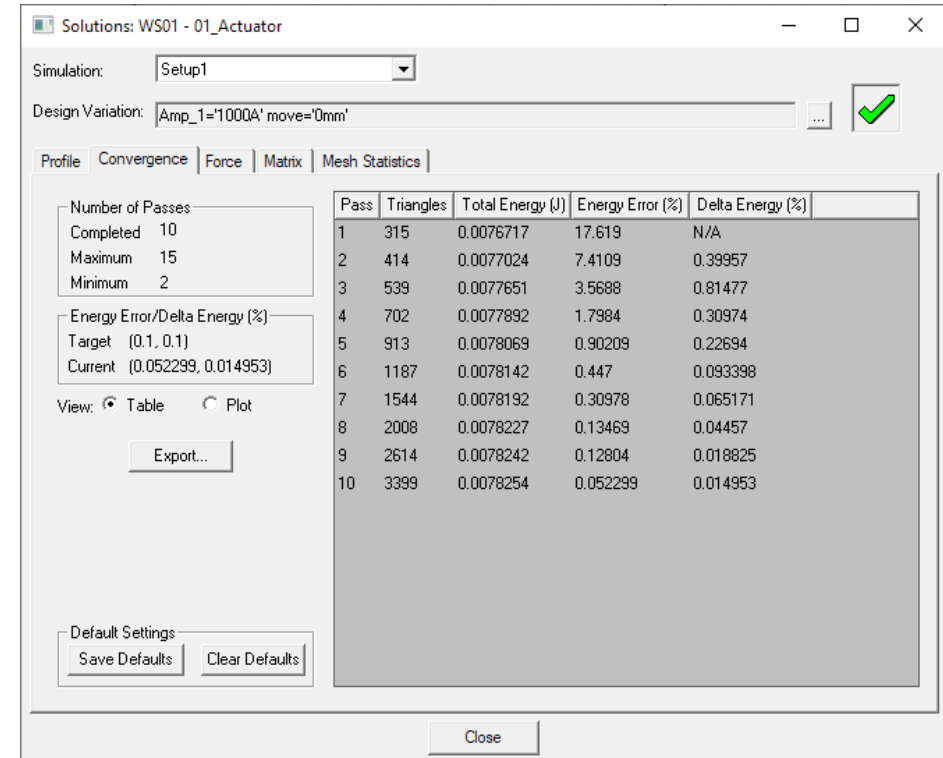
Analyze

- Create an analysis setup:
 - In Project Manager window *RMB on Analysis* → *Add Solution Setup*
 - Solution Setup Window:
 - **General Tab**
 - Maximum Number of Passes: 15
 - Percent error: 0.1%
 - **Solver Tab**
 - NonLinear Residual: 1e-6
 - Press OK
- Start the solution process:
 - In Project Manager window *RMB on Setup1* → *Analyze*



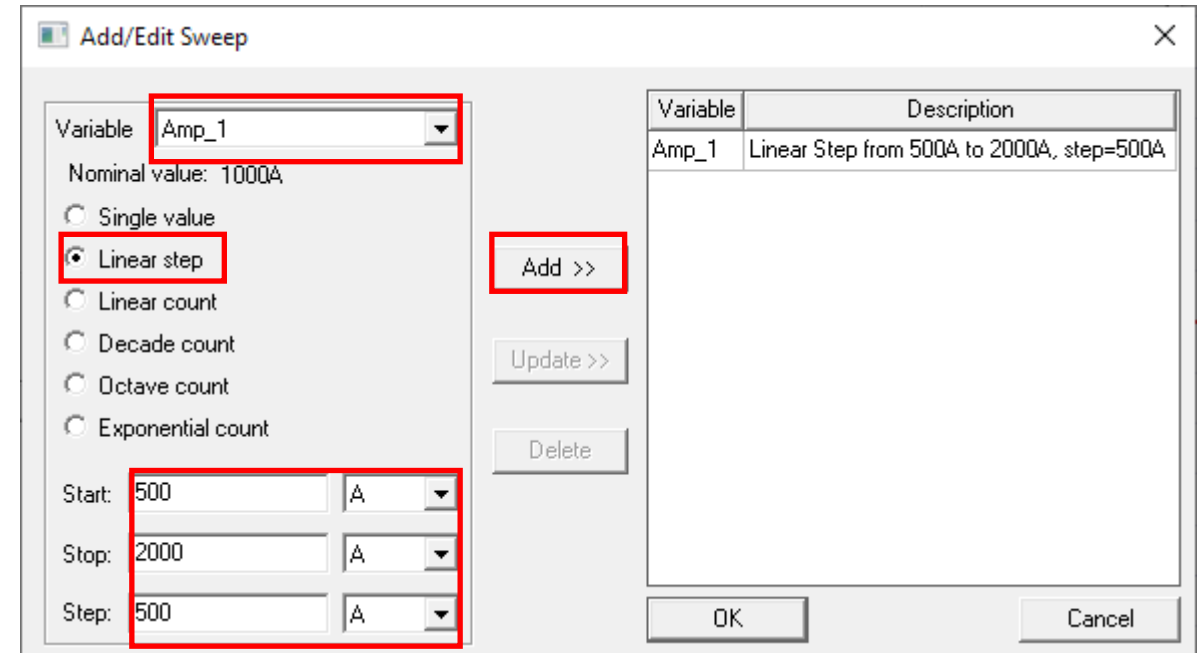
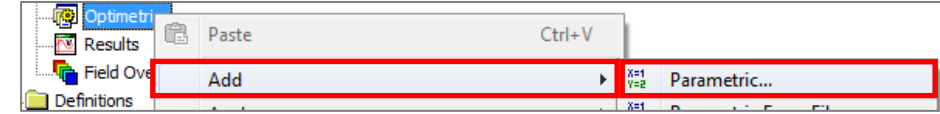
Solution Data

- View Solution Information
 - In Project Manager window **RMB on Results** → **Solution Data**
 - To View Convergence
 - Select the **Convergence tab**
 - To View Calculated Force Value
 - Select the **Force tab**



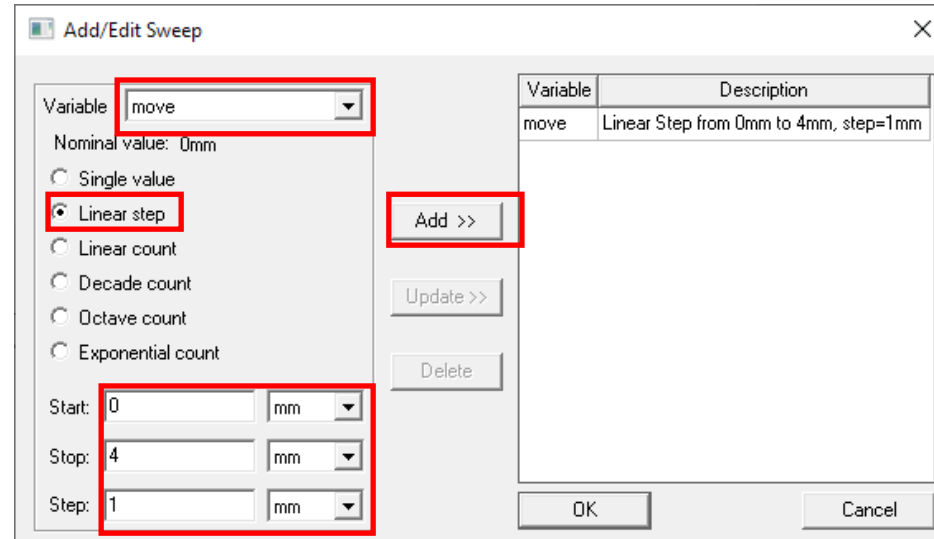
/ Set the Parametric Analysis

- Create a Parametric analysis setup:
 - In Project Manager window **RMB on Optimetrics** → **Add** → **Parametric**
 - Setup Sweep Analysis Window:
 - Press Add
 - Add/Edit Sweep Window:
 - Select Variable **Amp_1**
 - Linear Step: ☒ **Checked**
 - Start: 500 A
 - Stop: 2000 A
 - Step: 500 A
 - Press Add >>
 - Press OK



Set the Parametric Analysis

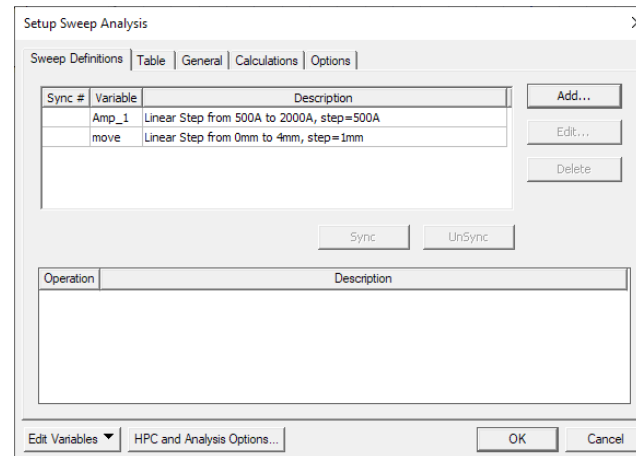
- Setup Sweep Analysis Window:
 - Press Add
- Add/Edit Sweep Window:
 - Select Variable **move**
 - Linear Step: ☒ Checked
 - Start: 0 mm
 - Stop: 4 mm
 - Step: 1 mm
 - Press Add >>
 - Press OK
- Setup Sweep Analysis Window:
- Options Tab
 - Save Fields And Mesh: ☒ Checked
 - Press OK
 - Press OK



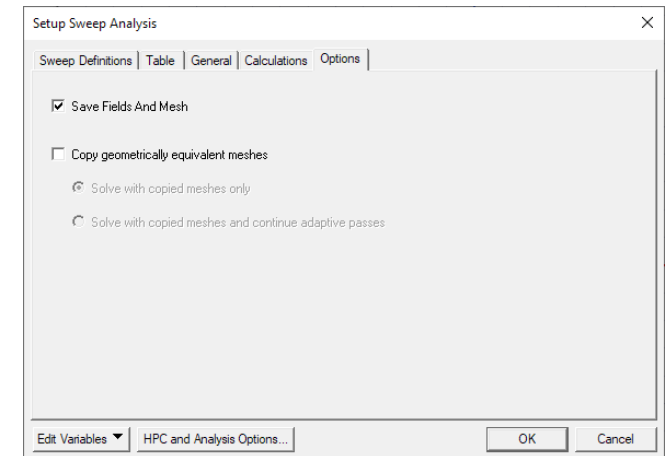
The 'Add/Edit Sweep' dialog box is shown. The 'Variable' dropdown is set to 'move'. The 'Nominal value' is '0mm'. The 'Linear step' radio button is selected. The 'Start' is '0 mm', 'Stop' is '4 mm', and 'Step' is '1 mm'. The 'Add >>' button is highlighted. The 'Description' field contains 'Linear Step from 0mm to 4mm, step=1mm'.

Setup Sweep Analysis

Sweep Definitions			Table	General	Calculations	Options
*	Amp_1	move				
1	500A	0mm				
2	500A	1mm				
3	500A	2mm				
4	500A	3mm				
5	500A	4mm				
6	1000A	0mm				
7	1000A	1mm				
8	1000A	2mm				
9	1000A	3mm				
10	1000A	4mm				
11	1500A	0mm				
12	1500A	1mm				
13	1500A	2mm				
14	1500A	3mm				
15	1500A	4mm				
16	2000A	0mm				
17	2000A	1mm				
18	2000A	2mm				
19	2000A	3mm				
20	2000A	4mm				



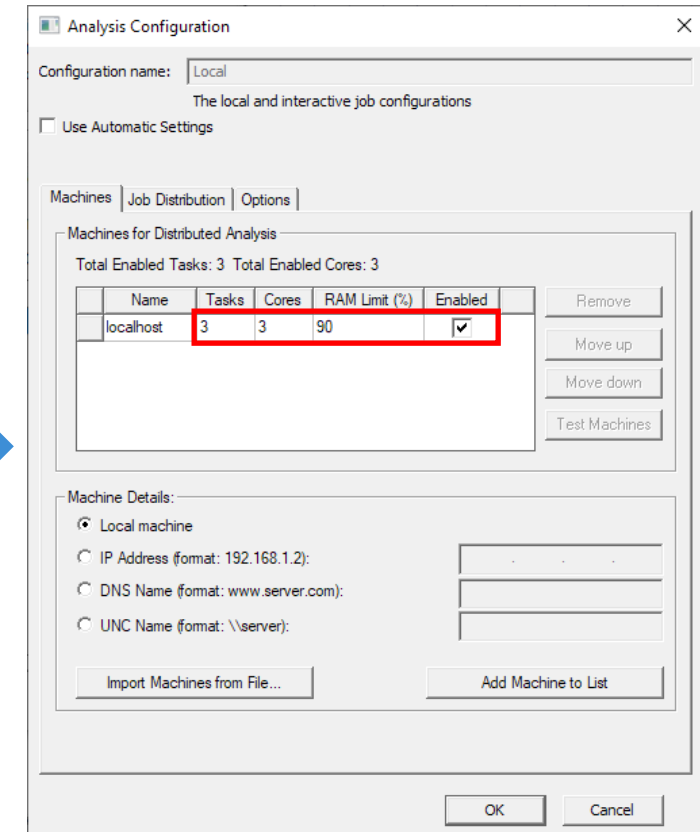
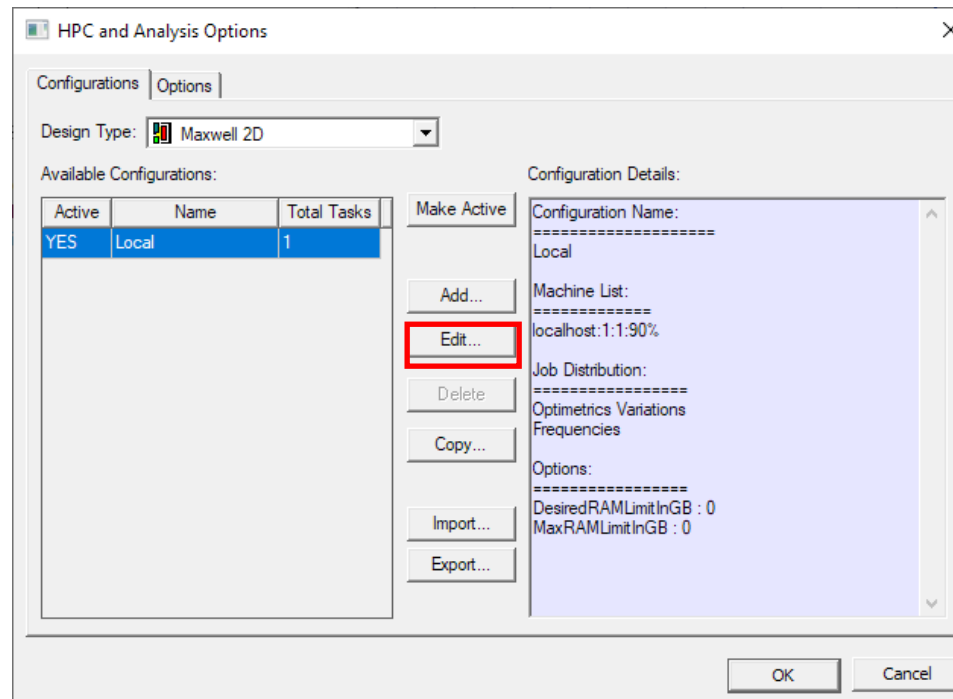
The 'Setup Sweep Analysis' dialog box is shown with the 'Sweep Definitions' tab selected. The table lists two sweeps: 'Amp_1' (Linear Step from 500A to 2000A, step=500A) and 'move' (Linear Step from 0mm to 4mm, step=1mm). The 'Add...', 'Edit...', and 'Delete' buttons are visible. The 'Sync' and 'UnSync' buttons are also present. The 'Operation' and 'Description' fields are empty.



The 'Setup Sweep Analysis' dialog box is shown with the 'Options' tab selected. The 'Save Fields And Mesh' checkbox is checked. The 'Copy geometrically equivalent meshes' checkbox is unchecked. The 'Solve with copied meshes only' radio button is selected. The 'Solve with copied meshes and continue adaptive passes' radio button is unselected. The 'Edit Variables' and 'HPC and Analysis Options...' buttons are visible. The 'OK' and 'Cancel' buttons are at the bottom.

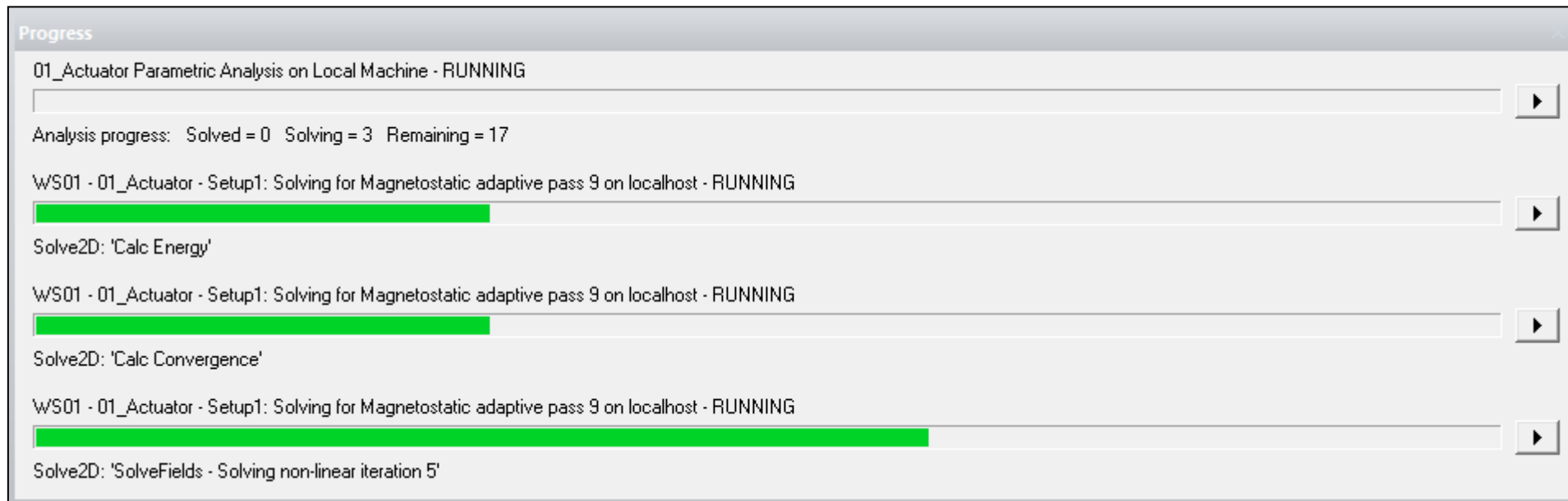
Settings DSO Analysis

- **Activate DSO (Distributed Solver Option)**
 - DSO is an additional option enabling the distribution of different sweeps through the processor's cores, speeding up Parametric analysis
 - Select menu **Tools** → **Options** → **HPC and Analysis Options**
 - **HPC and Analysis Options Window:**
 - Press **Edit**
 - **Tasks: 3**
 - **Cores: 3**
 - Press **OK**
 - Press **OK**



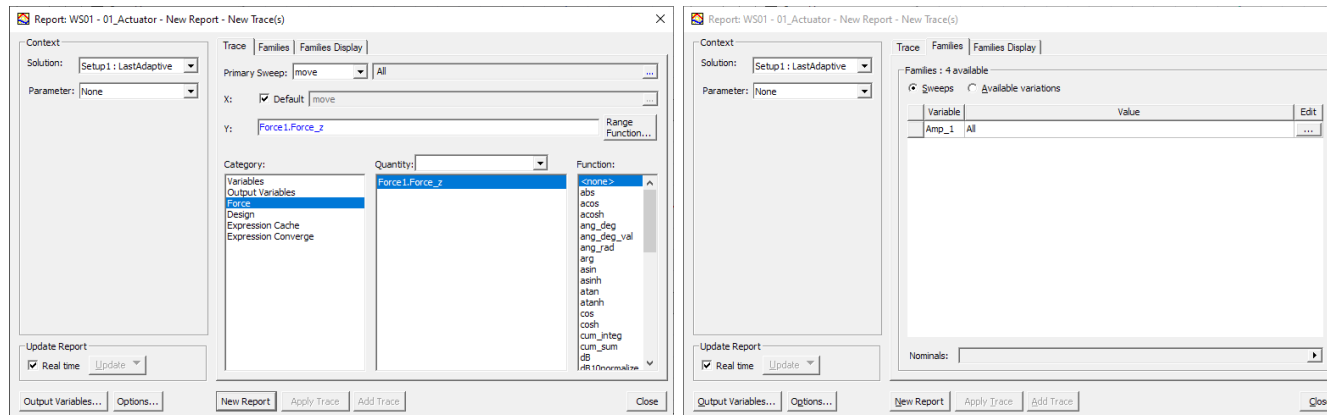
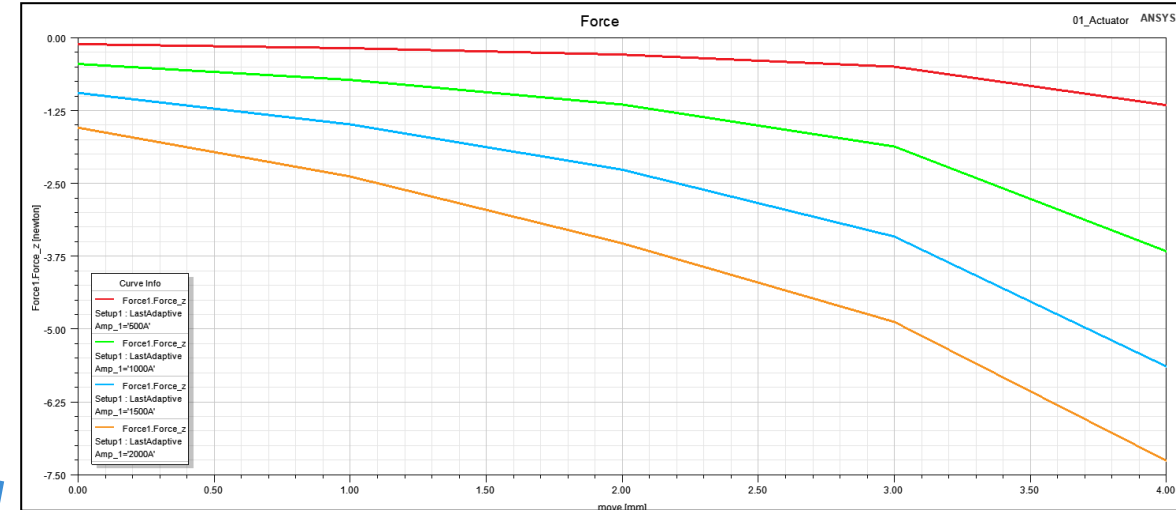
/ Analyze the Parametric Sweep

- Launch the Parametric Analysis
 - *RMB on ParametricSetup1 → Analyze*
 - In the Progress Window it is possible to see that 3 Sweeps at the same time are solved



Create XY Plot

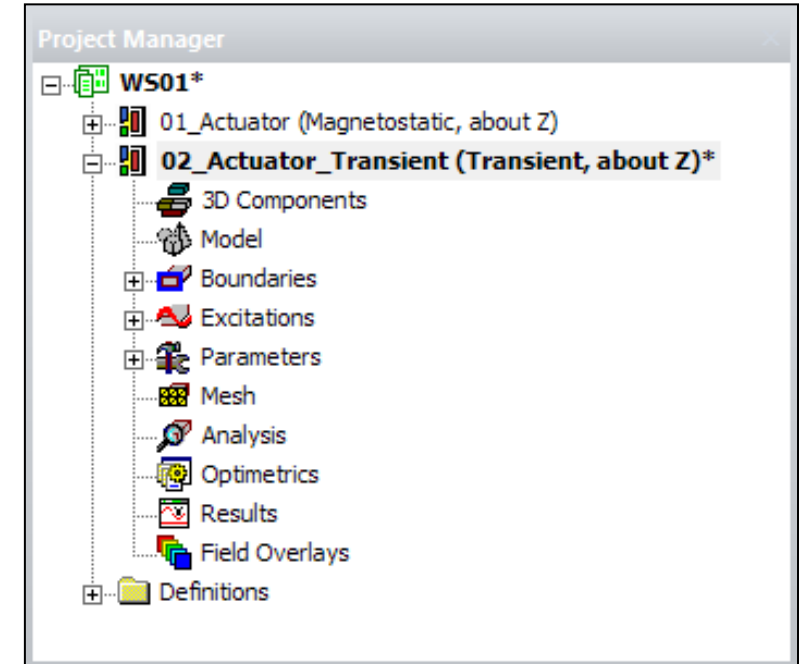
- Plot the values of Force depending on **Amp_1** and **move** Variables
 - Select the menu item **Maxwell 2D** → **Results** → **Create Magnetostatic Report** → **Rectangular Plot**
 - In Report window (**Trace Tab**)
 - Primary sweep:
 - X : **move**
 - Category: Force
 - Quantity: Force1.Force_z
 - **Families Tab**
 - **Amp_1: All**
 - **Select New Report**



Transient Model

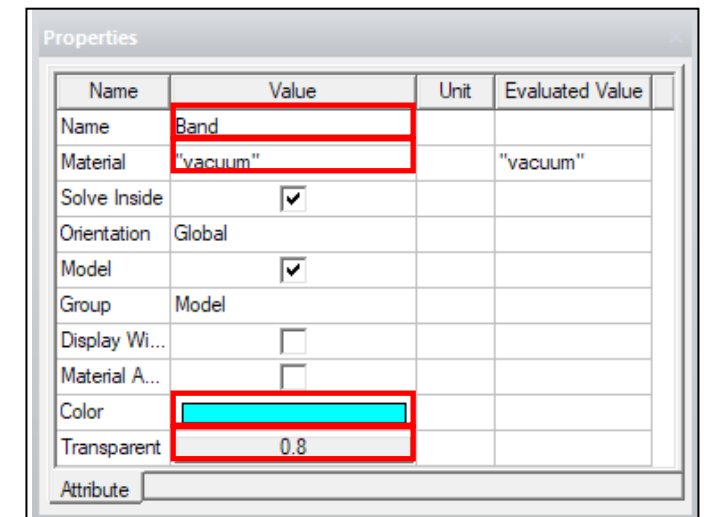
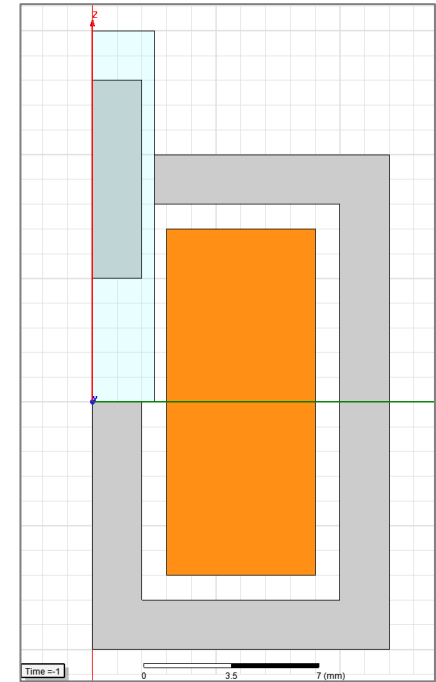
Create Transient Design

- Copy the 01_Actuator design
 - Select the Maxwell Design “01_Actuator”
 - Ctrl + C
 - Select the Maxwell Project
 - Ctrl + V
 - Click on the new Design name
 - Name the new Design “02_Actuator_Transient”
- Change the Solution Type
 - Select the menu item *Maxwell 2D → Solution Type*
 - Choose *Magnetic → Transient*
 - Click the OK button
- Delete the Parametric Analysis
 - Select and expand *Optimetrics*
 - Select ParametricSetup1, *RMB → Delete*



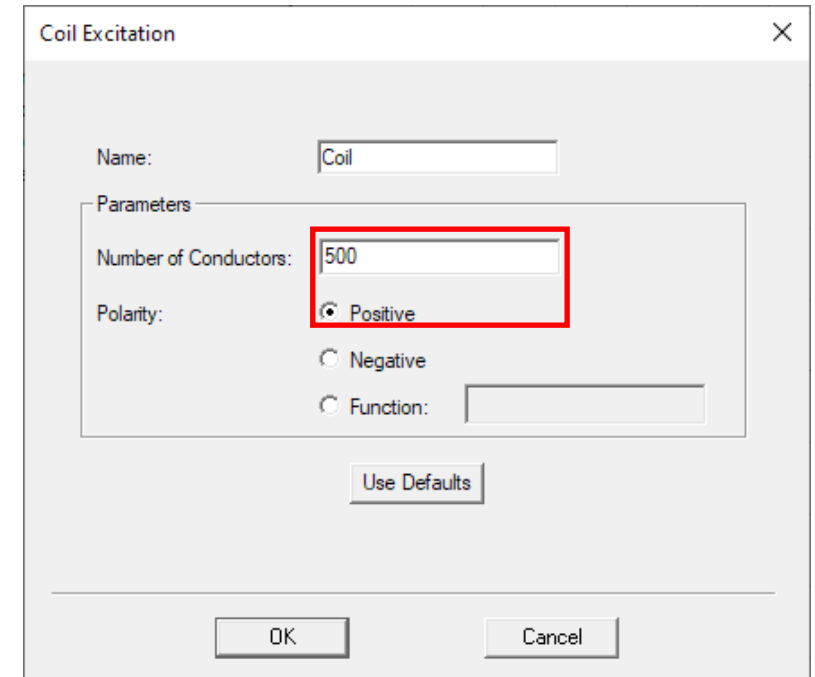
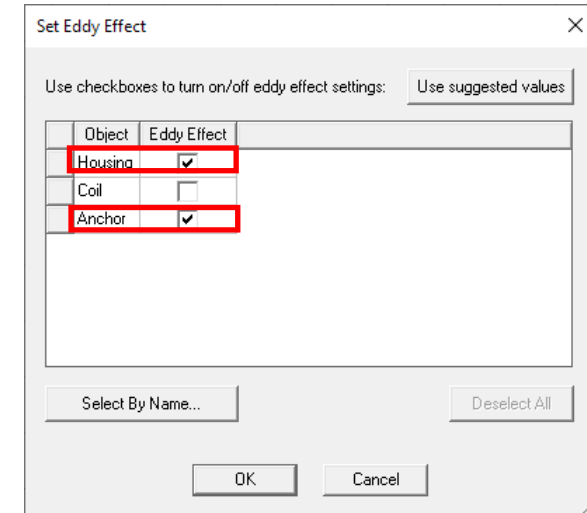
Create Band Object

- Create the Band
 - Select the menu item **Draw** → **Rectangle**
 - Using the coordinate entry fields, enter the position of rectangle
X: 0, Y: 0, Z: 15, *Press the Enter key*
 - Using the coordinate entry fields, enter the opposite corner
dX: 2.5, dY: 0, dZ: -15, *Press the Enter key*
 - Change the name of resulting sheet to **Band**, the color to **light Blue** and transparency to **0.8** using the property window
 - Leave the material as it is (Vacuum)



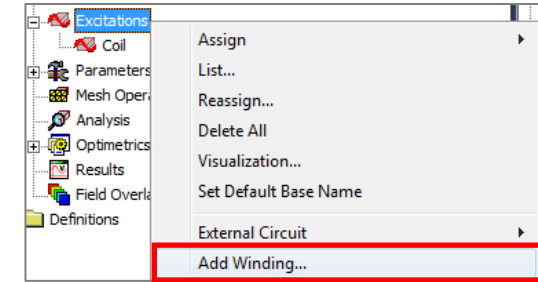
Modify Excitation

- **Cancel Existing Excitation**
 - Select the **Current1** under Excitation and delete it
- **Set Eddy Effects**
 - Select menu item **Maxwell 2D → Excitations → Set Eddy Effects**
 - Housing: ☒ Checked
 - Anchor: ☒ Checked
- **Assign new Excitation**
 - Select the sheet **Coil**
 - Select menu item **Maxwell 2D → Excitations → Assign → Coil**
 - In Coil Excitation window,
 - Number of Conductors: 500
 - Positive: ☒ Checked
 - Press OK



Modify Excitation

- **Modify Excitation – Create windings**
 - Select menu item *Maxwell 2D* → *Excitations* → *Add winding*
 - Name: Winding1
 - Type: Voltage
 - Stranded: ☒ Checked
 - Resistance: 2 Ω
 - Voltage: 10 V
 - Press OK
- **Add Coils to Windings**
 - *RMB on Winding1* → *Add Coils*
 - Choose **Coil**
 - Press OK



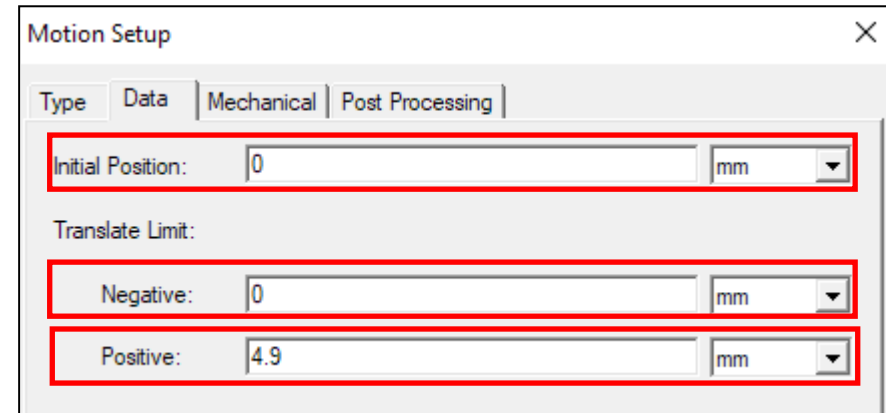
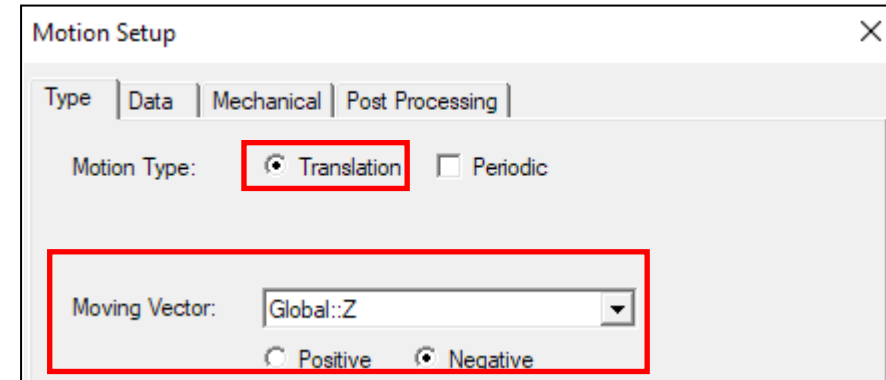
The 'Winding' dialog box is shown. The 'Name' field contains 'Winding1'. The 'Type' is set to 'Voltage'. The 'Stranded' radio button is selected. The 'Resistance' is set to '2 ohm'. The 'Voltage' is set to '10 V'. The 'Number of parallel branches' is set to '1'. The 'Use Defaults' button is visible at the bottom.

The 'Add Terminals' dialog box is shown. The 'Terminal Listing Options' section has 'Terminals not assigned to this winding' selected. The table below shows a single entry: 'Coil' with 'Conductor Number' 500 and 'Currently Assigned To' empty.

Coil Terminal	Conductor Number	Currently Assigned To
Coil	500	

Assign Motion

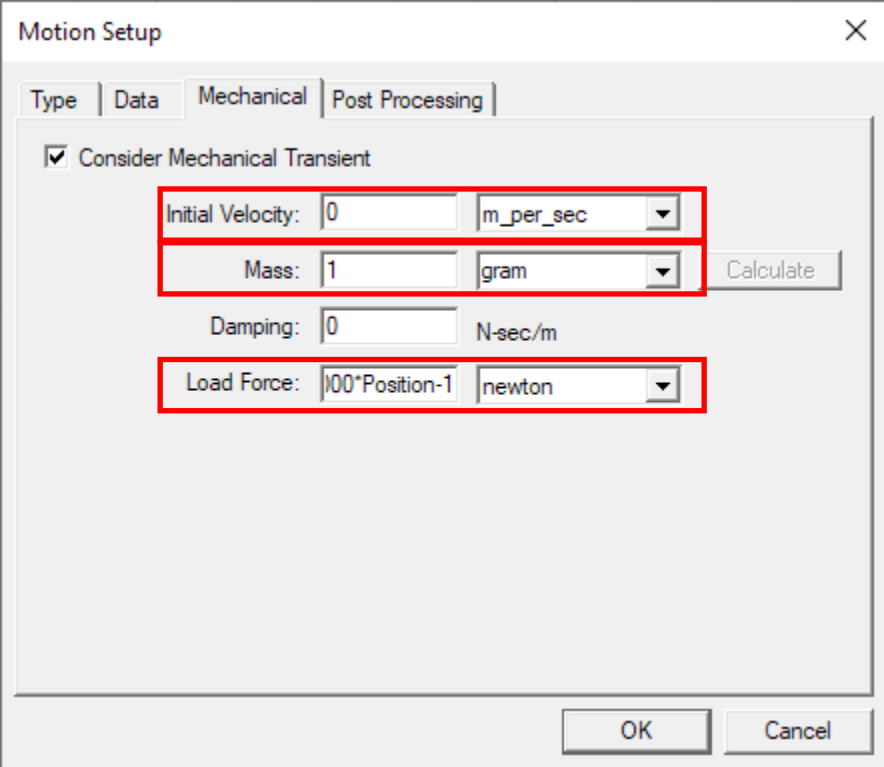
- Specify motion
 - Select the sheet **Band** from history tree
 - *Select menu item Maxwell 2D → Model → Motion setup → Assign Band*
 - In Motion Setup window,
 - *Type tab*
 - Motion Type: Translation
 - Moving Vector: Global:Z
 - Negative: ☒ Checked
 - *Data tab*
 - Initial Position: 0 mm
 - Negative: 0 mm
 - Positive: 4.9 mm



Note: Initial position of 0 mm indicates that motion will start at $t = 0$ with the Anchor position being as in the geometry. A non-zero initial position (P_0) indicates the simulation would start with Anchor moved by P_0 from the current position

/ Assign Motion

- *Mechanical tab*
- Consider Mechanical transient: ☒ Checked
- Initial Velocity: 0
- Mass: 1 gram
- Damping: 0 N-sec/m
- Load Force: $-1000 \cdot \text{Position} - 1$
- Press OK



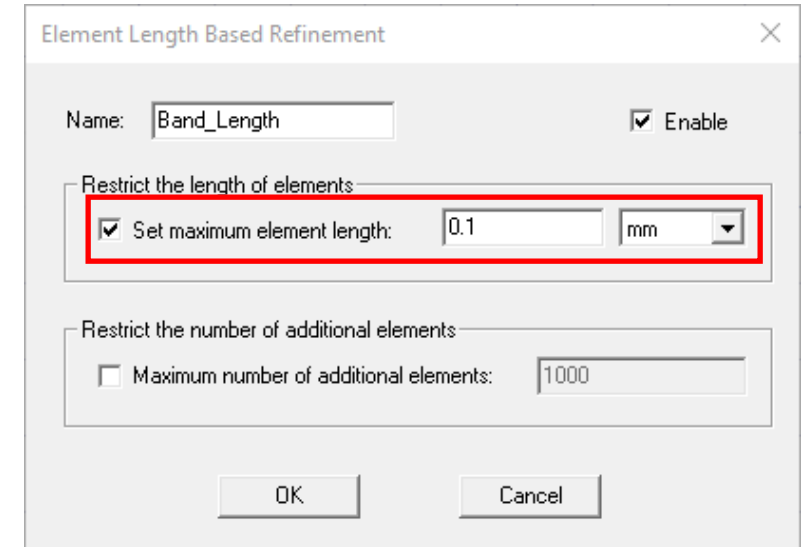
The screenshot shows the 'Motion Setup' dialog box with the 'Mechanical' tab selected. The 'Consider Mechanical Transient' checkbox is checked. The following parameters are configured:

- Initial Velocity: 0 m_per_sec
- Mass: 1 gram
- Damping: 0 N-sec/m
- Load Force: $-1000 \cdot \text{Position} - 1$ newton

A 'Calculate' button is located to the right of the Mass field. At the bottom right, there are 'OK' and 'Cancel' buttons.

/ Mesh Operations

- Assign Mesh Operations on Band
 - Select the object **Band** from the history tree
 - Select menu item **Maxwell 2D → Mesh Operations → Assign → Inside Selection → Length Based**
 - In Element Length Based Refinement window,
 - Name: Band_Length
 - Set maximum element length: ☒ **Checked**
 - Set maximum element length: 0.1 mm
 - Maximum number of additional elements: ☐ **Unchecked**
 - Press OK



Element Length Based Refinement

Name: Band_Length ☒ Enable

Restrict the length of elements:

☒ Set maximum element length: 0.1 mm

Restrict the number of additional elements:

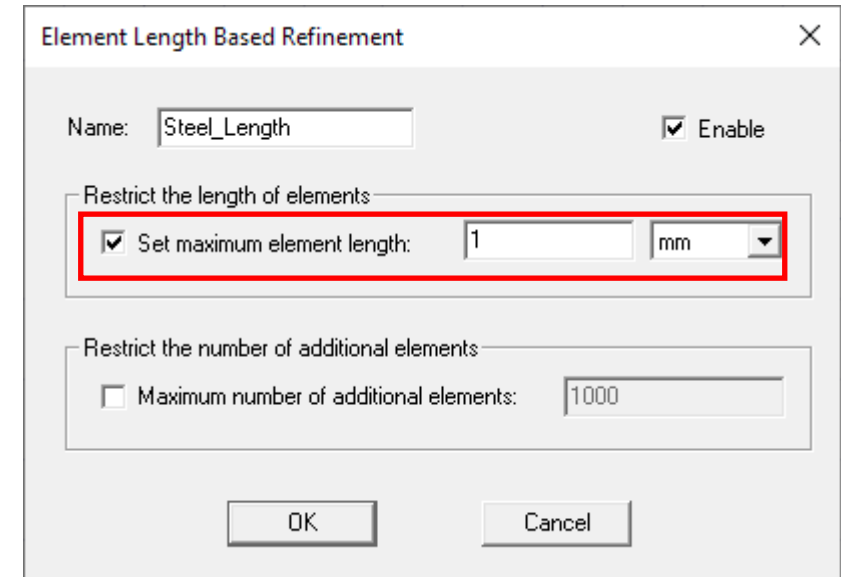
☐ Maximum number of additional elements: 1000

OK Cancel

Note: Meshing is a very critical issue with respect to simulation speed and accuracy. For Force computation, the most critical area is the airgap. Thus, the band mesh is crucial for accurate results.

/ Mesh Operations

- Assign Mesh Operations on Housing and Anchor
 - Select both the sheets **Housing** and **Anchor** using the Ctrl key
 - Select menu item **Maxwell 2D → Mesh Operations → Assign → Inside Selection → Length Based**
 - In Element Length Based Refinement window,
 - Name: Steel_Length
 - Set maximum element length: ☒ **Checked**
 - Set maximum element length: 1 mm
 - Maximum number of additional elements: ☐ **Unchecked**
 - Press OK



Element Length Based Refinement

Name: Steel_Length ☒ Enable

Restrict the length of elements

☒ Set maximum element length: 1 mm

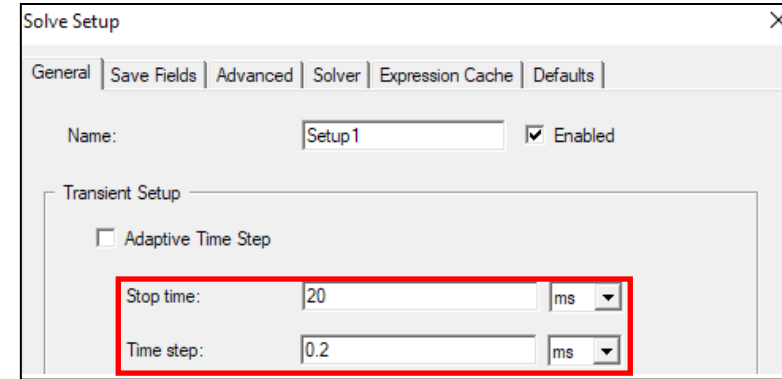
Restrict the number of additional elements

☐ Maximum number of additional elements: 1000

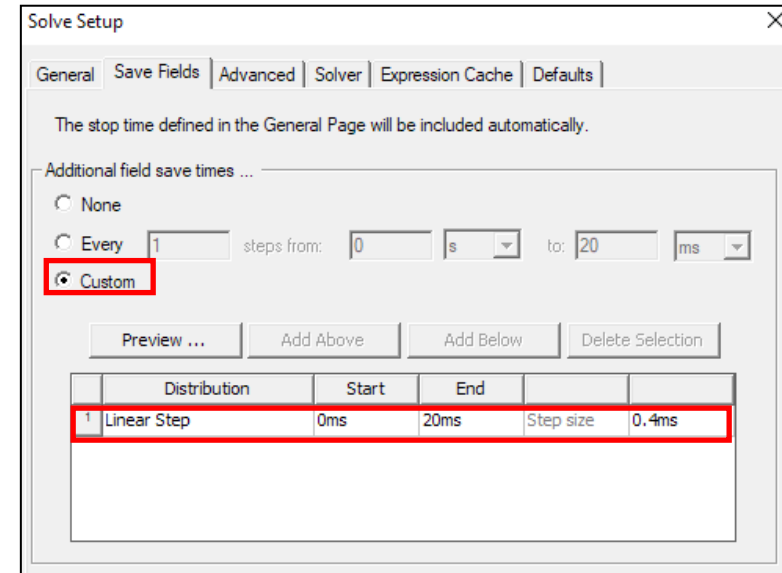
OK Cancel

Create Analysis Setup

- Create Analysis Setup
 - Select menu item *Maxwell 2D* → *Analysis Setup* → *Add Solution Setup*
 - In Solve Setup window,
 - General Tab
 - Stop Time: 20 ms
 - Time Step: 0.2 ms
 - Save Fields Tab
 - Select Custom
 - Type: linear Step
 - Start: 0 ms
 - Stop: 20 ms
 - Step: 0.4 ms
 - Press Add to list >>
 - SolverTab
 - NonLinear Residual: 1e-6
 - Press OK



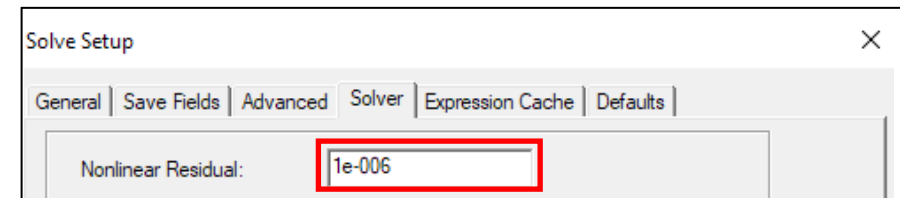
The screenshot shows the 'Solve Setup' dialog box with the 'General' tab selected. The 'Name' field is set to 'Setup1' and the 'Enabled' checkbox is checked. Under the 'Transient Setup' section, the 'Adaptive Time Step' checkbox is unchecked. The 'Stop time' is set to 20 ms and the 'Time step' is set to 0.2 ms. These two fields are highlighted with a red rectangle.



The screenshot shows the 'Solve Setup' dialog box with the 'Save Fields' tab selected. A message states: 'The stop time defined in the General Page will be included automatically.' Under 'Additional field save times ...', the 'Custom' radio button is selected and highlighted with a red rectangle. Below this, there are buttons for 'Preview ...', 'Add Above', 'Add Below', and 'Delete Selection'. A table lists the saved field times:

	Distribution	Start	End		
1	Linear Step	0ms	20ms	Step size	0.4ms

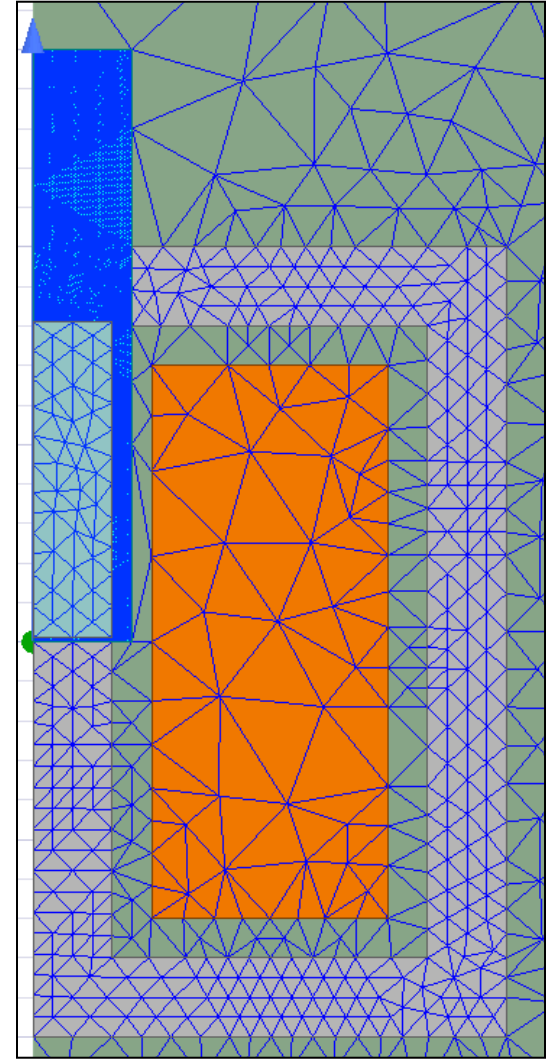
The first row of the table is highlighted with a red rectangle.



The screenshot shows the 'Solve Setup' dialog box with the 'Solver' tab selected. The 'Nonlinear Residual' is set to 1e-006, which is highlighted with a red rectangle.

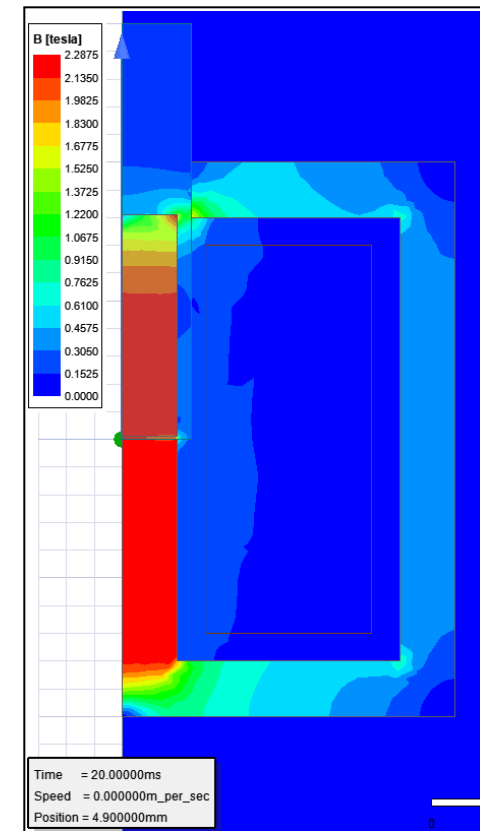
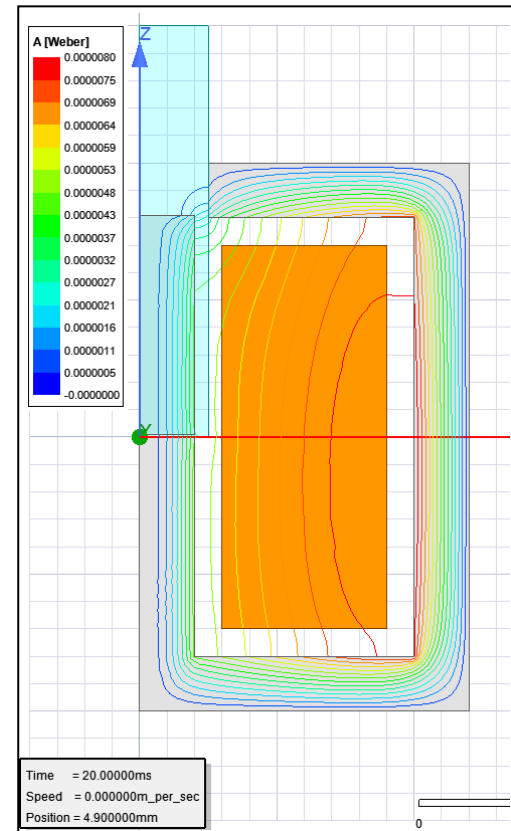
/ Analyze and Plot Mesh

- Run the Solution
 - In Project Manager window *RMB on Setup1 → Analyze*
- Plot Mesh
 - Select the menu item *View → Set Solution Context*
 - In Set view Context window,
 - Change Time to 0.02 s
 - Press OK
 - Select the menu item *Edit → Select All (Ctrl+A)*
 - Select the menu item *Maxwell 2D → Fields → Plot Mesh*



Results – Create Fields Plot

- Plot Flux_lines
 - Select the menu item *Edit* → *Select All*
 - *Maxwell 2D* → *Fields* → *Fields* → *A* → *Flux_Lines*
 - In Create Field Plot window,
 - Press Done
- Plot Mag_B
 - Select the menu item *Edit* → *Select All*
 - *Maxwell 2D* → *Fields* → *B* → *Mag_B*
 - In Create Field Plot window,
 - Press Done



Note: To hide the previously created field plots, right click on plot from Project manager window and uncheck “Plot Visibility”

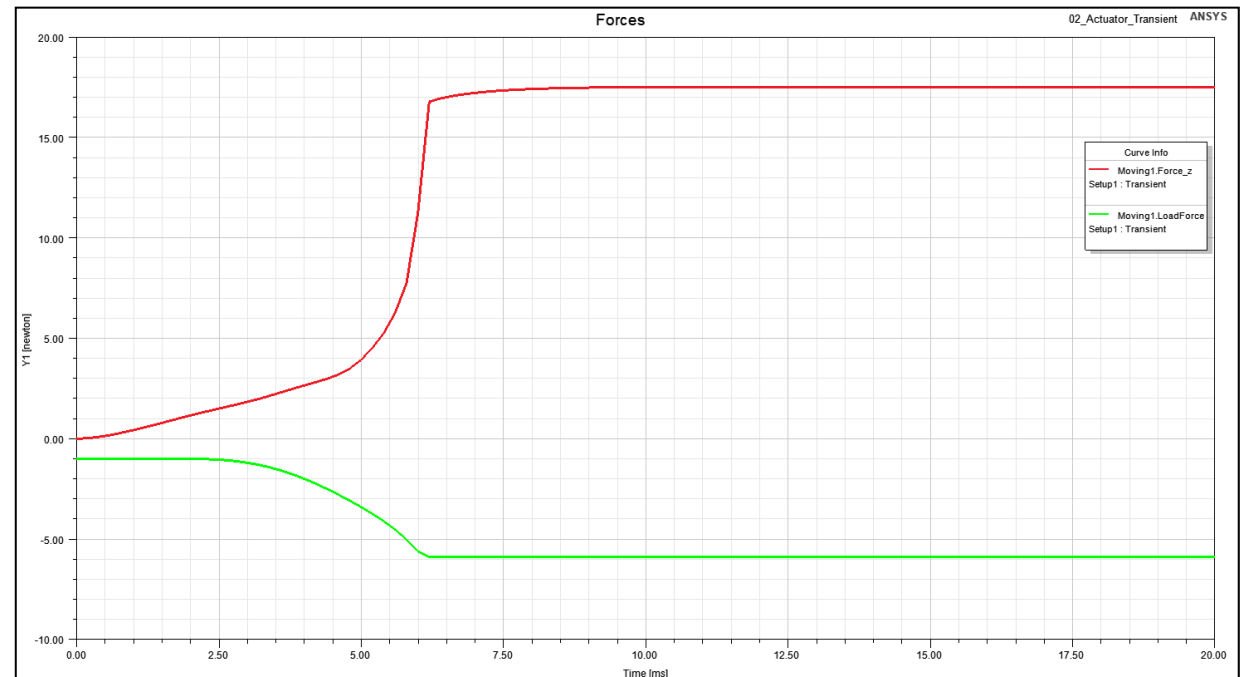
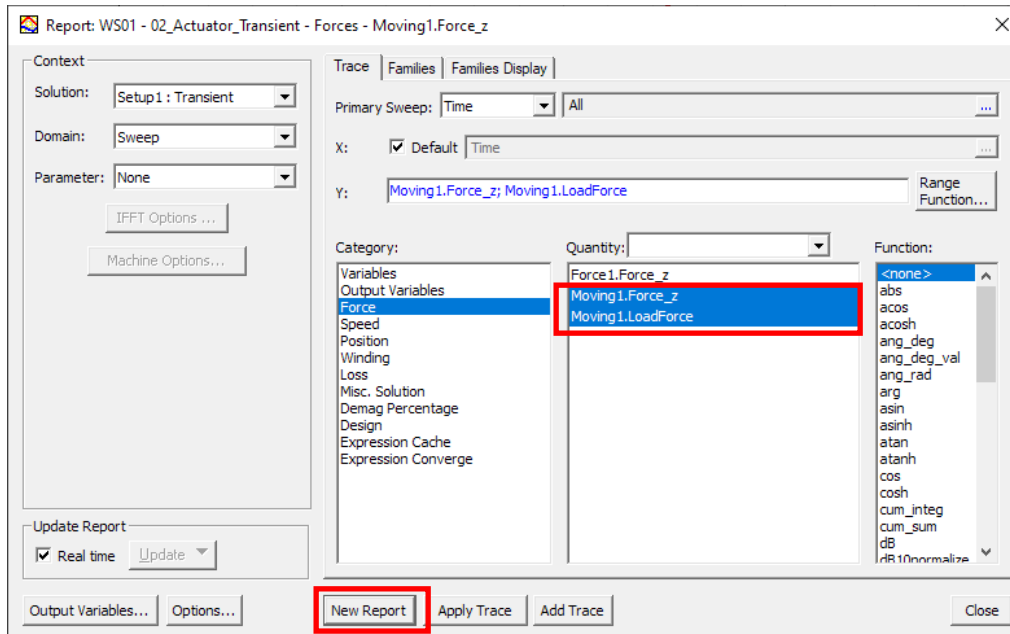
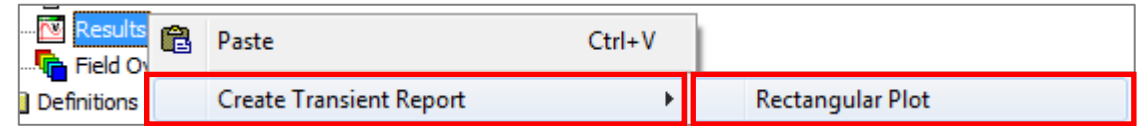
Plot Forces vs. Time

- Create a XY Plot

- In Project Manager window *RMB on Results* → *Create Transient Reports* → *Rectangular Plot*

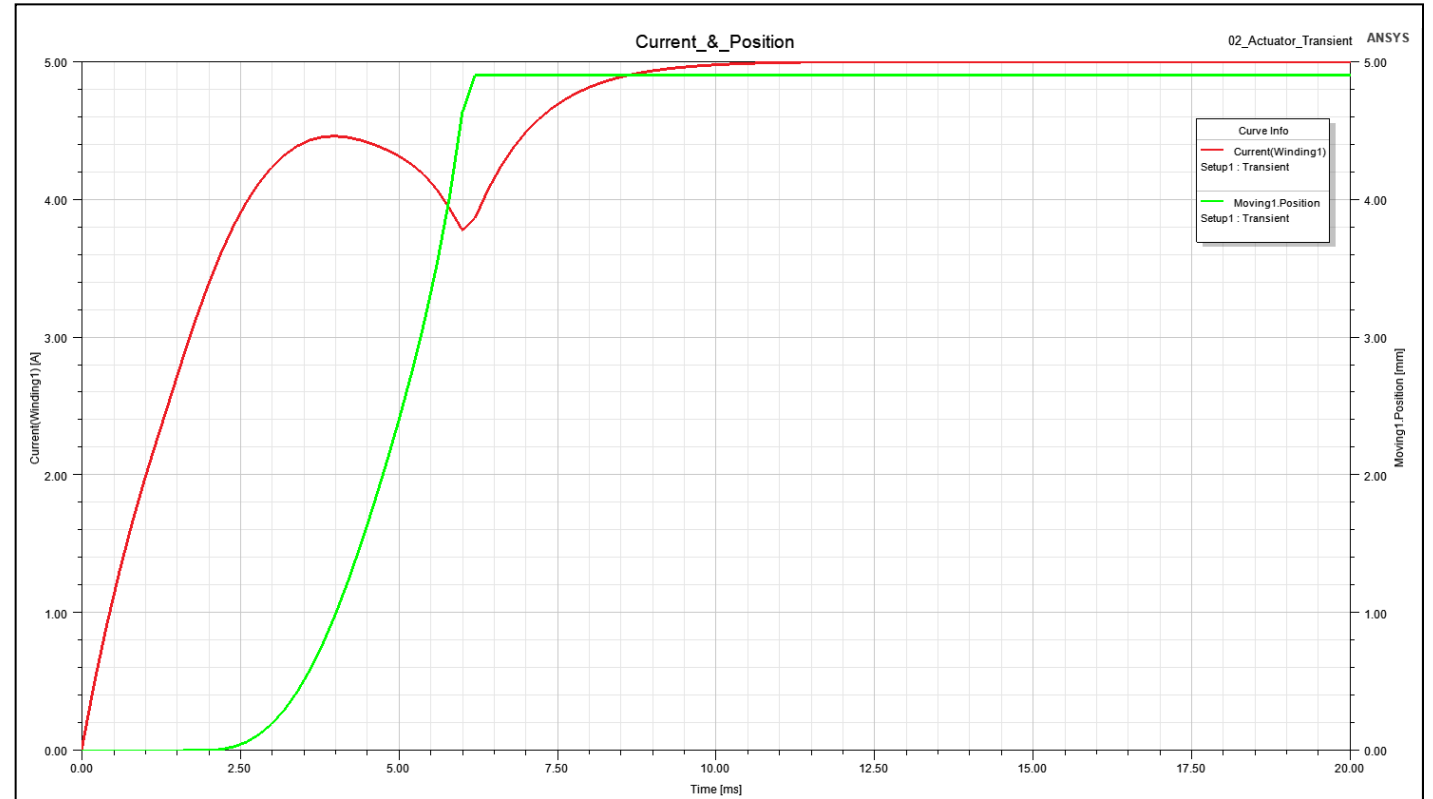
- In Reports window,

- Category: Force
 - Quantity: Moving1.Force
 - Quantity: Moving1.LoadForce
 - Select New Report



Plot Current and Position vs. Time

- Create a XY Plot
 - In Project Manager window *RMB on Results* → *Create Transient Reports* → *Rectangular Plot*
 - In Reports window,
 - Category: Winding
 - Quantity: Current(Winding1)
 - Select New Report
 - Category: Position
 - Quantity: Moving1.Position
 - Add Trace
 - Close



Saving the Project

- This completes the workshop
- Save the file with the name **Workshop 01** in the working folder